

NYCU NLP Lab

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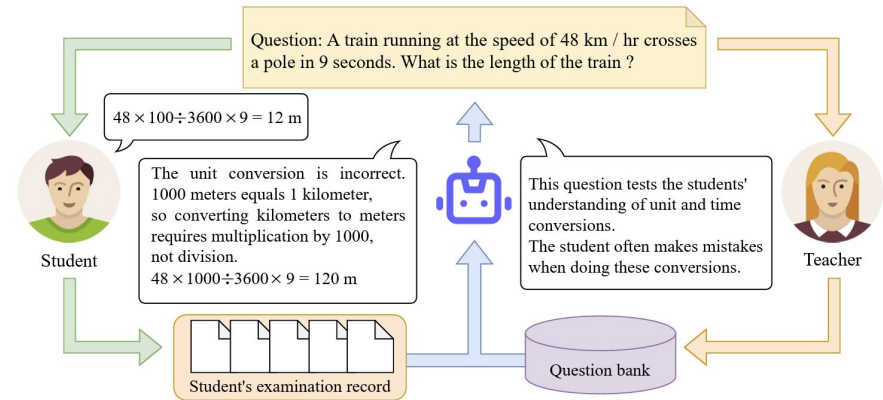
Outline

- Research Achievements
- Ongoing Research Topics
- Lab FAQ

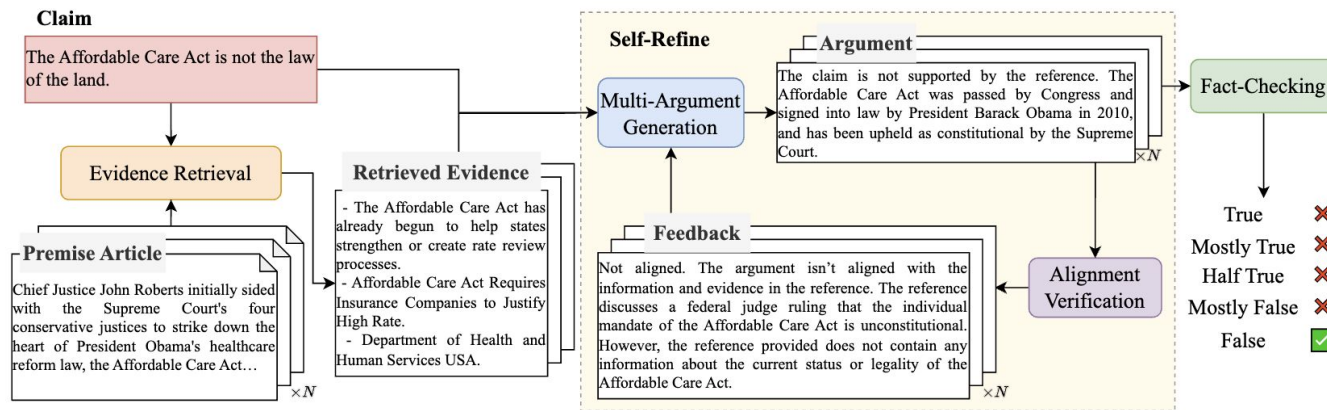
Research Achievements

LLMs and the Future of Mathematics Education (EMNLP 2023 Findings)

- Explores LLMs' ability to provide adaptive feedback in math learning
- Tests LLMs' problem-solving and error correction on MathQA
- 66% accuracy; providing rationales misleads model (only 10% correctly identifies correct answers), struggles with mathematical equations



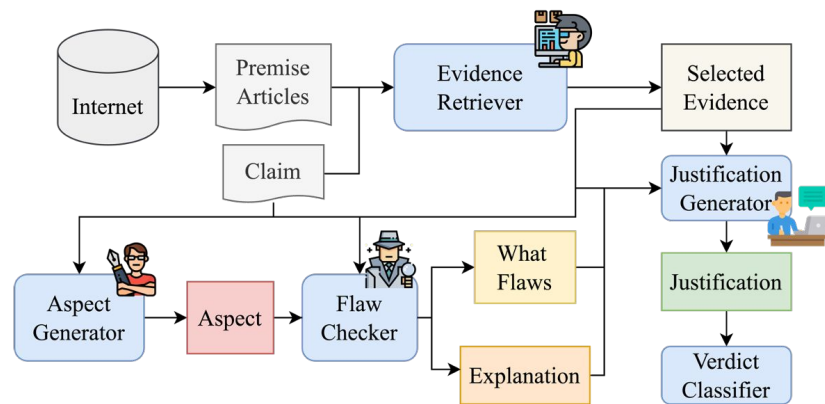
Cross-Domain Fact-Checking (COLING 2024)



- Cross-domain fact-checking faces evidence extraction and verification challenges
- MAGIC uses multi-argument generation and self-refinement to reconstruct evidence
- Performs well on out-of-domain and partially true/false claims; self-refinement effective

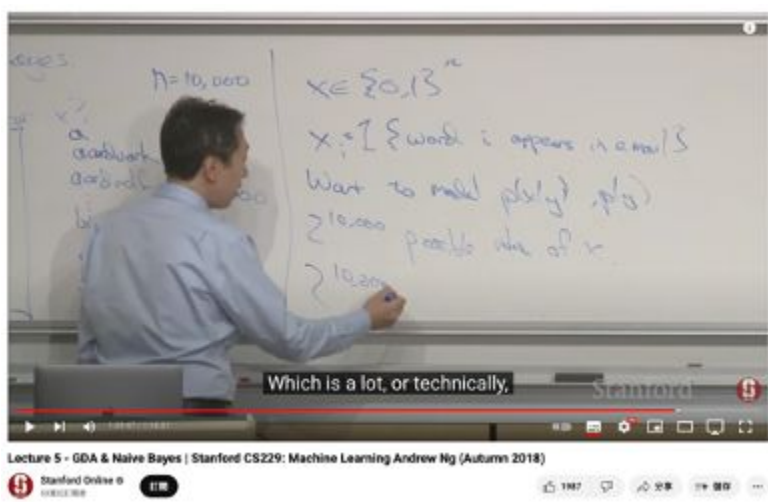
How We Refute Claims: Automatic Fact-Checking through Flaw Identification and Explanation (WWW 2024)

- Existing fact-checking lacks flaw identification and deep explanation for false claims
- We propose RefuteClaim framework defining 7 flaw types with aspect and justification generation
- Our model performs well in classifying and explaining false claims; best with all flaws considered



Course Content Question Forecasting

Given the course content, is it possible to predict the questions students may ask?



@kevinsiao1148 2 個月前
great lecture! but anyone knows at 1:09:43 why we need 2^10000 parameters here? Thanks a lot!

@humanity_first48 6 個月前
can any one help me out with this question " why we only use standard gaussian distribution?"

@vineethparsi1625 1 個月前
At 1:09:48 why do we need 2power10000 parameters i think its 2*10000 because each dimension needs two parameters and variance is same anyways.

Based on the content provided about the Naive Bayes algorithm and GDA, students might have the following questions:

1. Can you explain again why there are $2^{10,000}$ possible values of x ?
2. What do you mean by " x is a binary vector of this 10,000 dimensional"?
3. Why do we need $2^{10,000} - 1$ parameters instead of $2^{10,000}$?
4. Can you elaborate on the assumption of conditional independence in the Naive Bayes algorithm?
5. Why do we use the chain rule of probability in this context?
6. How does the assumption of conditional independence simplify our calculations?
7. Is there a reason for choosing the Naive Bayes algorithm over other classifiers for this particular problem?
8. What would be the challenges if we didn't make the assumption of conditional independence?
9. How does the Naive Bayes algorithm handle overfitting given such a large number of dimensions?
10. If x is binary, does that mean our dataset consists only of 0s and 1s?
11. What's the practical significance of modeling $P(x|y)$ and $P(y)$ in a generative learning algorithm?
12. How does the Naive Bayes algorithm compare with GDA in terms of performance and use cases?
13. How do we handle cases where there might be a zero frequency problem, leading to zero probabilities?
14. Can the conditional independence assumption be problematic or introduce biases in any way?
15. How would we handle the situation if some of the features are not binary?

Remember, these are potential questions students might ask based on the content given. Some might ask deeper or more advanced questions depending on their prior knowledge or understanding of the topic.

@vineethparsi1625 9 個月前
At 1:09:48 why do we need 2power10000 parameters i think its 2*10000 because each dimension needs two parameters and variance is same anyways.

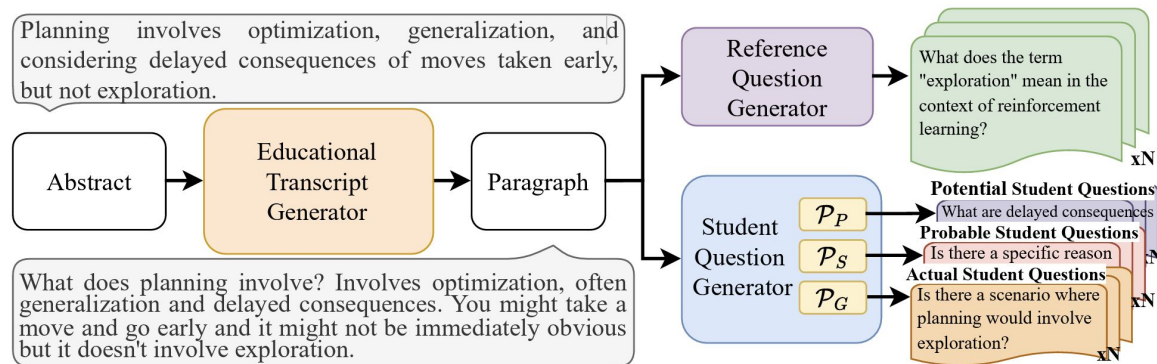
2 則回覆

1 則回覆

@hariprakash1715 4 個月前 (已編輯)
because each have two possibilities and there 2power 10000 means there are those many possibilities of feature vector

1 則回覆

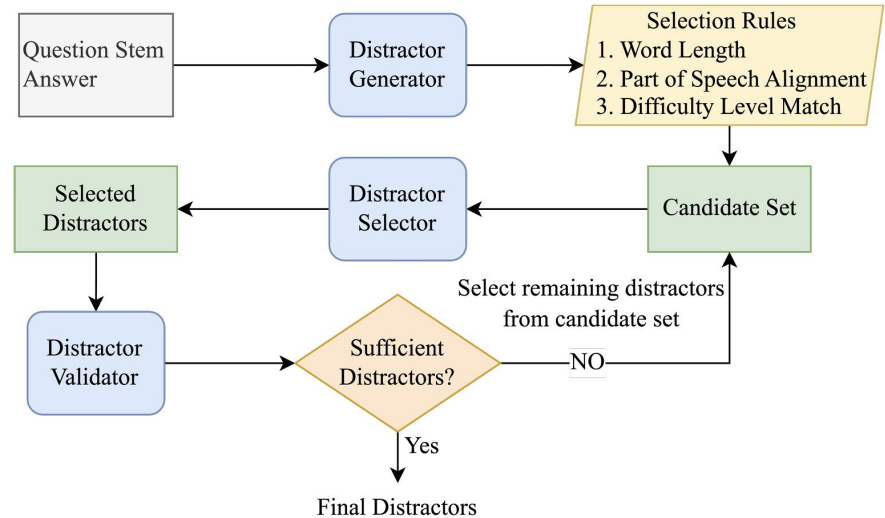
Educational Lecture Abstract-based Question Generation System (IJCAI 2024 Demo)



- The need for anticipating student questions helps reduce lecture preparation effort
- The E-QGen system uses lecture abstracts with multitask LoRA-tuned LLMs to generate student-like question
- E-QGen achieves higher similarity and diversity than GPT-4

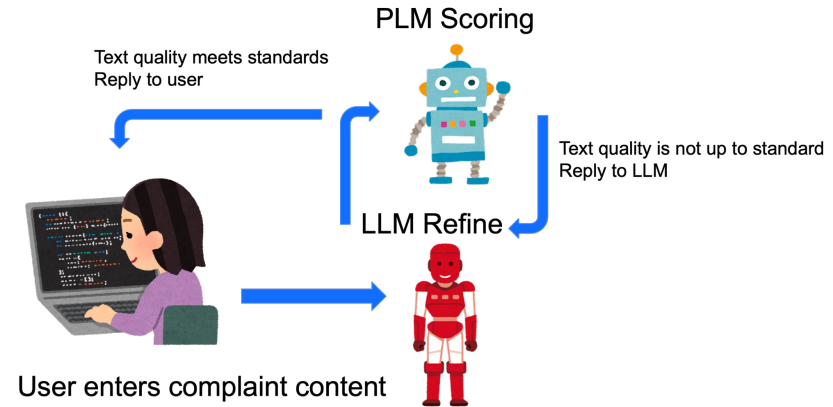
Vocabulary Test Distractor Generation (ACML 2025)

- Distractor design is time-consuming and existing methods generate invalid options
- ISSR uses LLM iterative selection with self-review to avoid multiple correct answers
- Self-review effectively filters invalid options, challenges advanced learners



Refining Financial Consumer Complaints

- Financial consumer complaints often lack legal formality and persuasiveness
- The MSMI framework combines a classifier and LLMs for iterative legal text refinement
- MSMI significantly outperforms single-pass generation in success rate and robustness



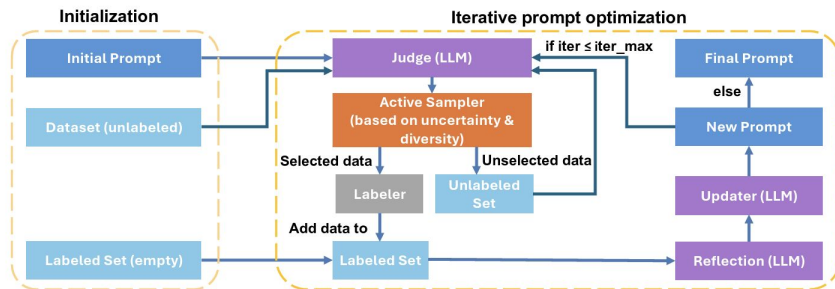
Ongoing Research Topics

Effective Design of LLM Judges

How to Build Effective LLM Judges with Minimal Human Cost?

- Minimal human annotation
- Automatic evaluation guideline construction
- LLM confidence assessment

Related Work

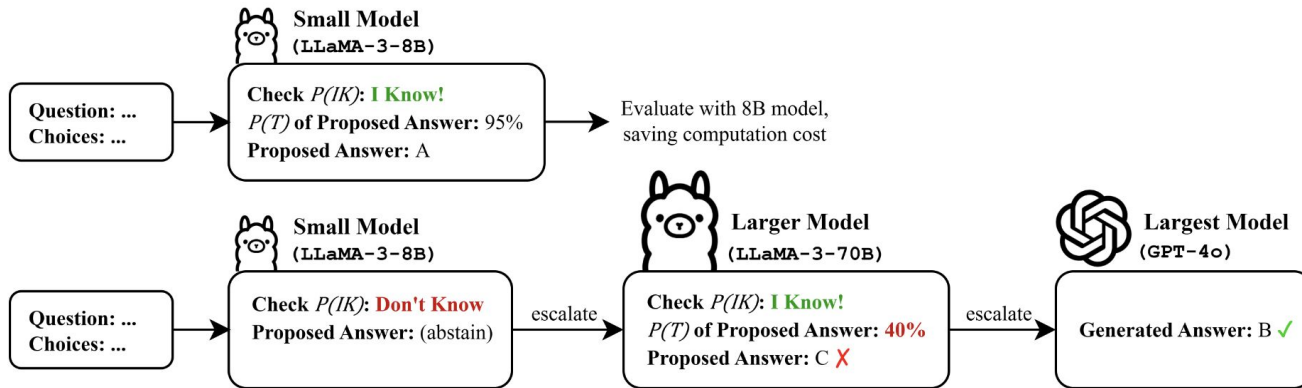


<https://aclanthology.org/2025.acl-industry.67.pdf>

Confidence-Driven Multi-Scale Model Selection

As LLMs continue to scale, always using the largest model for inference becomes cost-prohibitive, motivating methods that balance accuracy and computational cost

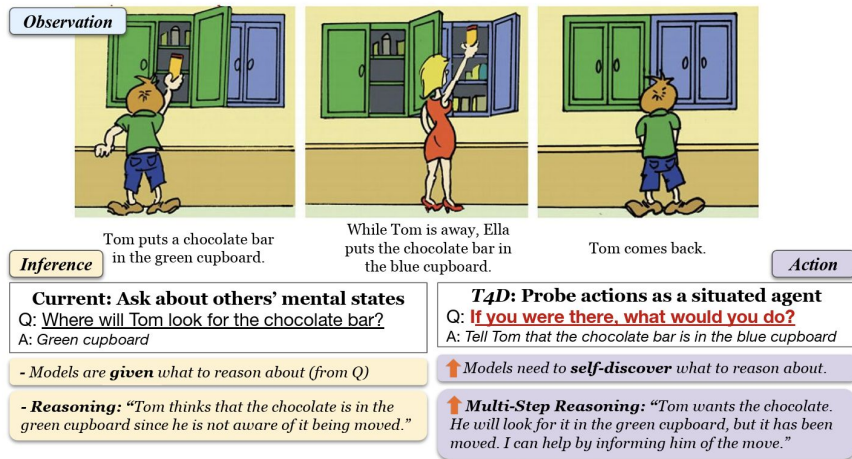
How can LLM confidence estimates be used to reduce inference cost while maintaining accuracy?



Task-Oriented Theory-of-Mind

- How to enable agents to accomplish given tasks in simulated environments (e.g., what to pick up, where to go)?
- How far has research progressed on hallucination issues in planning tasks?

Related Work



<https://arxiv.org/pdf/2310.03051>

Is Generated Content Unsafe?

Current image moderation focuses on explicit harms but fails to capture subtle affective and unsettling visual risks in real and AI-generated images

This motivates a systematic evaluation of VLMs' ability to understand affect, context, and visual discomfort



Safe?



Safe?

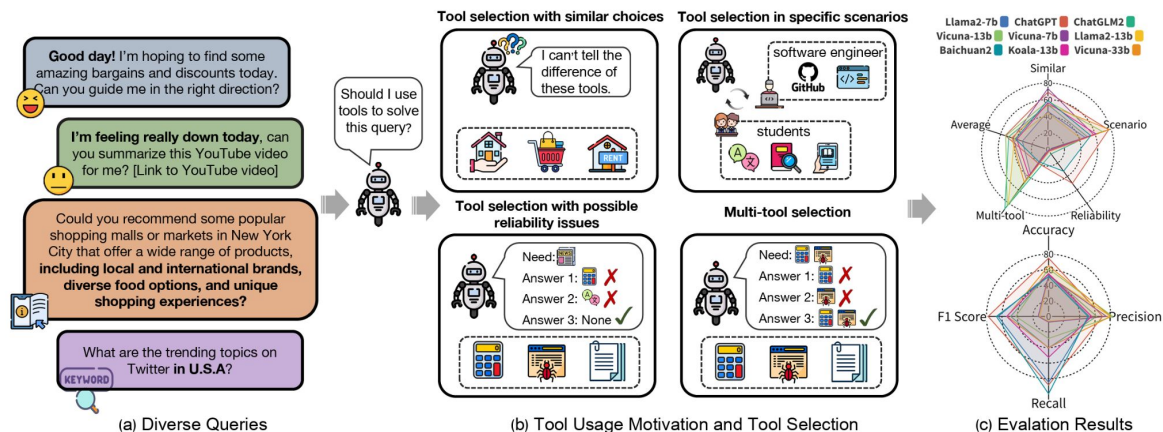


Safe?

Tool Calling

Can LLMs autonomously decide when to use tools, and correctly select among multiple available tools?

Related Work



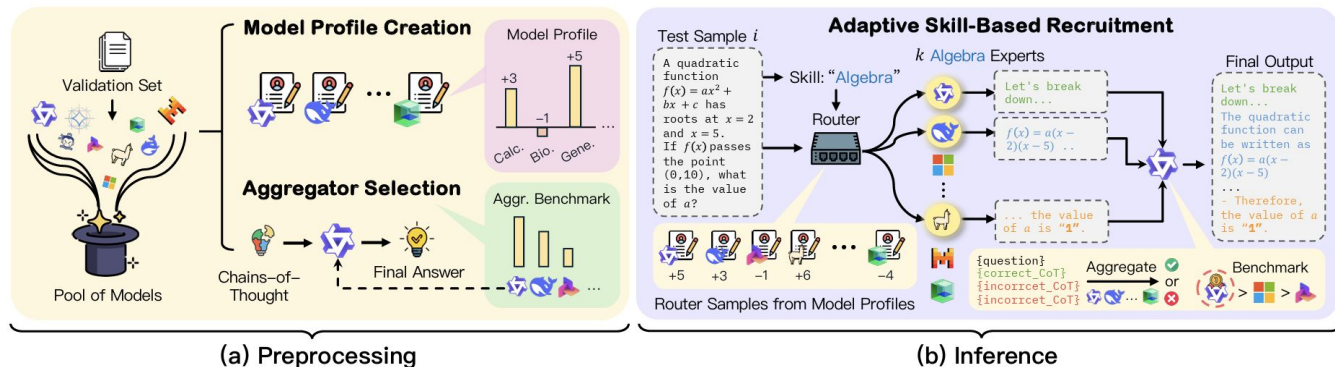
<https://arxiv.org/pdf/2310.03128>

Routing

As LLMs vary widely in capability and cost, routing all queries to large models is not cost-effective in practice

This motivates routing queries to appropriate LLMs to balance output quality and inference cost

Related Work



Towards Adaptive Feedback for Student Mathematical Problem-Solving

- Evaluating LLMs on real student problem-solving processes
- LLM capability for providing adaptive feedback
- Collection of real student solutions with teacher annotations

Problem: Two trains of equal length are running on parallel lines in the same directions at 46 km/hr and 36 km/hr. The faster train passes the slower train in 144 seconds. The length of each train is:

Student Process:

$$46 - 36 = 10$$

$$10 \text{ km/hr} \times \frac{5}{18} = \frac{100}{36} \text{ m/s}$$

$$\frac{100}{36} \times 144 = 400 \text{ m}$$

Error Type: Wrong Mathematical Operation/Concept

Error Equation:

$$\frac{100}{36} \times 144 = 400 \text{ m}$$

Teacher Feedback: To overtake the other train, you need to travel the combined length of both trains. Since both trains are of the same length, you need to divide by 2 to get the answer.

Lab FAQ

FAQ

問題1: 實驗室需要值班嗎？

答: 不需要，但是實驗室提供每位同學固定座位。

問題2: 實驗室有國科會或是業界的計畫嗎？

答: 實驗室有正在執行的國科會計畫與產學合作計畫。同學的研究津貼也從這些計劃提供。

問題3: 需要自己找研究題目，還是老師會指定？

答: 我會提供研究主題，如果同學有自己感興趣的研究題目，也可以和我討論。

FAQ

問題4: 實驗室是否有固定的seminar或個人meeting?

答: 每週安排一次個人meeting, 跟我討論研究進度, 會議以實體方式進行為主。另外, 每週也安排一次seminar, 由同學輪流報告論文, 學習論文閱讀、報告、和回答技巧。

問題5: 實驗室運算資源?

答: 實驗室有支援高速運算與實驗之機架式工作站級設備, 可以處理大量的資料及運算, 進行大規模實驗。

問題6: 平均都幾年畢業?

答: 碩士生基本畢業時間為2年。

問題7: 需要學生擔任助教?

答: 需要。